



LAB -4-: Image Classification Using Deep Features

November 2, 2025

1. Objective:

- Extract deep features from images using two pre-trained models, VGG16 and ResNet, and use these features for image classification to compare their performance.

2. Programming Language and Library

- UC Merced Land Use Dataset.
- Python with OpenCV and scikit-learn libraries.
- TensorFlow: For deep learning models.

3. Lab Procedure:

- Data Loading and Preprocessing:
- Dataset Splitting:
 - Split the dataset into a training set and a testing set.
 - Ensure that the split maintains a balance of images from different categories.
- **Deep Feature Extraction** : For each image in the dataset:
 - Steps to Extract Deep Features:
 - Load VGG16 model without the final classification layer.
 - Process the image through the model to obtain feature descriptors.
 - Extract the output from the last fully connected layer.
 - Store the extracted features for each image.
- Classifier Training:
 - Train a classifier (Support Vector Machine) using the extracted deep features and the corresponding class labels.
- Generate Predictions:
 - For each test image, extract features, and use the trained classifier to predict the class.
- Performance Evaluation:
 - Compute accuracy to evaluate the SVM classifier's performance.
 - Draw and include the confusion matrix in your lab report.

4. Assignment: Provide a report on the outcomes.